CASCADIA ANALYTICS

Taiwan Semiconductor Manufacturing Co

Equity Research Note | TSM | October 29, 2026

Priced Beyond Perfection as N2 Capital Intensity Weighs on Free Cash Flow Conversion

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Taiwan Semiconductor Manufacturing Co maintains an unassailable technological monopoly in leading-edge logic, delivering unprecedented top-line expansion driven by high-performance computing accelerators and the rapid adoption of the 2-nanometer node. However, the equity market has decoupled from the fundamental reality of extreme capital intensity, rising depreciation dilution, and structural overseas cost premiums that compress free cash flow conversion. Even under constructive operational assumptions, the current market capitalisation of $2,244.5B discounts multi-year compounding that exceeds cash flow realization. We initiate coverage with a Hold rating and valuation concerns, establishing a probability-weighted blended target of $284.15, implying -34.3% upside versus the spot price of $432.75.

HOLD	$284.15	-34.3%	$432.75
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The broader market remains fixated on record quarterly top-line prints and accelerating high-performance computing mix, extrapolating perpetual multiple expansion from top-line beats. Our quantitative framework demonstrates that front-loaded capital expenditures of $60-64 billion in 2026 and margin compression from

early N2 depreciation create an inescapable multi-year drag on terminal cash flow yields.

Executive summary

Taiwan Semiconductor Manufacturing Co is the world's preeminent pure-play semiconductor foundry, manufacturing leading-edge logic for fabless designers including NVIDIA, AMD, Apple, and Broadcom. The company's commercial revenue is heavily concentrated in advanced nodes, where 7-nanometer and smaller architectures account for 79% of total wafer revenue. High-performance computing platform revenue has expanded rapidly to comprise 68% of the corporate mix, reinforcing the company's status as the monopoly supplier of sub-5nm logic integrated with CoWoS advanced packaging solutions. This manufacturing nexus positions the foundry at the center of the global artificial intelligence hardware buildout.

The evidentiary record over recent fiscal quarters underscores extraordinary commercial demand matched with intensifying capital demands. In Q3 FY2026, the company printed record revenue of $46.72 billion, representing 41.2% year-over-year growth and exceeding the top end of guidance, while net profit reached NT$763.06 billion, up 68.7% year over year. The 2-nanometer node demonstrated unprecedented customer migration velocity, surging from 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026. However, this transition initiated early-stage N2 depreciation that compressed gross margin from 67.7% in Q2 to 66.4% in Q3, while cumulative nine-month capital expenditures climbed to $40.64 billion. This followed a strong Q2 FY2026, where revenue reached $40.20 billion and operating margin hit 60.3%, prompting management to raise the full-year 2026 capex forecast to $60-64 billion and the full-year revenue growth outlook to more than 40%.

From a balance sheet perspective, the enterprise maintains exceptional structural liquidity alongside high reinvestment requirements. In FY2025, the company generated $3,809.1B in revenue, $2,281.3B in gross profit (59.9% margin), $1,936.1B in operating profit (50.8% margin), and $1,697.6B in net income, delivering an ROE of 31.5%. Free cash flow reached $992.4B, translating to an FCF margin of 26.1%. The balance sheet closed FY2025 with $2,767.9B in cash and $1,064.6B in total debt, leaving a strong net cash reserve of $1,703.3B. Nevertheless, with 2026 capex elevated to $60-64 billion, capital allocation remains predominantly dedicated to massive fab buildouts and advanced node deployment, constraining structural cash conversion relative to peak cyclical earnings.

Operational snapshot

$432.75	$3,809.1B	$992.4B	26.1%	1.37	$432.75
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Absolute and Monopolistic

Taiwan Semiconductor Manufacturing Co possesses a wide economic moat rooted in insurmountable switching costs and unmatched process integration. As the monopoly supplier of sub-5nm logic with CoWoS advanced packaging, the company's fabless customers, including NVIDIA, AMD, Apple, and Broadcom, face severe technical and commercial friction should they attempt to migrate designs to alternative foundries. The co-design process for complex high-performance computing accelerators requires years of physical layout optimization, custom intellectual property libraries, and proprietary packaging qualification. A defection from TSMC's manufacturing ecosystem would introduce multi-quarter product delays and unacceptable yield penalties in leading-edge AI deployment.

The company's commercial pricing power is validated by historical margin performance and node adoption trends. Across FY2025, gross margin reached 59.9% before surging through FY2026 to 66.2% in Q1 and 67.7% in Q2. In Q4 FY2025, gross margin cleared guidance by up to 3.3 percentage points, printing at 62.3% as strong utilization and node pricing offset heavy investment. Fabless designers lack viable commercial alternatives at 3nm and 2nm, granting the foundry significant leeway to pass through escalating tool costs and wafer fabrication expenses, even as the broader hardware supply chain faces macroeconomic tightening.

Customer retention is reinforced by the ecosystem depth of TSMC's advanced packaging platforms. The integration of CoWoS advanced packaging with sub-5nm silicon has created an architectural standard for high-bandwidth accelerator clusters. Because advanced packaging capacity remains tightly paired with front-end wafer fabrication, fabless customers must commit multi-year volume allocations to secure CoWoS capacity alongside advanced wafers. This dynamic was demonstrated in Q1 FY2026 when management raised the 2024-2029 AI accelerator revenue CAGR outlook to 54-56% from 45%, tying the upgrade directly to hyperscaler buildout commitments.

However, the technological barrier of node transitions introduces structural operational challenges. The transition from 3nm, which accounted for 25% of revenue in Q1 FY2026 and 28% in Q4 FY2025, to 2nm introduces substantial front-loaded equipment depreciation. Gross margin contracted to 66.4% in Q3 FY2026 as N2 scaled to 8% of wafer revenue, and Q4 FY2026 gross margin was guided to 65-67%. Furthermore, overseas fab expansions carry structural cost premiums relative to Taiwan operations, while working capital requirements have expanded, with inventory days rising to 87 from 76 and receivable days climbing to 29 from 23 during Q2 FY2026.

Valuation

Three scenarios make the valuation range explicit

Cascadia Analytics evaluates Taiwan Semiconductor Manufacturing Co using an explicit, five-year discounted cash flow framework covering FY2026 through FY2030. Our modeling establishes that while the enterprise generates elite operational margins, the reference market capitalization of $2,244.5B at a spot price of $432.75 reflects an unsustainable premium over structural cash generation capabilities. The model incorporates a weighted average cost of capital of 9.79%, an equity beta input of 1.37, a base terminal growth rate of 3.0%, and diluted share count of 5.19 billion.

The valuation bridges three distinct operating scenarios across revenue growth and free cash flow conversion. In our Base Case, revenue compounds at an 11.6% CAGR over the forecast window, with terminal free cash flow margin reaching 23.0%, generating a per-share equity value of $282.08. In the Bear Case, revenue compounds at a 9.0% CAGR with terminal FCF margin compressing to 20.9%, resulting in a per-share target of $199.67. In the Bull Case, revenue compounds at a 15.5% CAGR with FCF margin sustaining at 23.5%, yielding a target of $372.79. Weighting these outcomes 25% Bear, 50% Base, and 25% Bull yields a blended target of $284.15, implying -34.3% upside versus the spot price.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	20.9%	$199.67	-53.9%
Base	50%	11.6%	23.0%	$282.08	-34.8%
Bull	25%	15.5%	23.5%	$372.79	-13.9%

Bear $199.67

Reflects an outcome where geopolitical friction, including 25% Section 232 tariffs and shifting export controls, restricts addressable demand while structural overseas cost premiums and prolonged N2 margin drag compress terminal FCF margin to 20.9% on a 9.0% revenue CAGR.

Base $282.08

Models steady high-performance computing expansion with an 11.6% revenue CAGR through FY2030, where capital expenditure intensity normalizes following the initial 2nm buildout and target FCF margin settles at 23.0%.

Bull $372.79

Captures sustained hyperscaler accelerator investment achieving a 15.5% revenue CAGR, rapid yields on the 2nm ramp that overcome depreciation drag, and disciplined capital allocation preserving a 23.5% terminal FCF margin.

Valuation framework

A live discount rate, not a static hurdle

Our quantitative framework relies on a dynamic 9.79% WACC that incorporates an equity beta input of 1.37 across 5.19 billion diluted shares. This hurdle rate captures the heightened operational and geopolitical volatility inherent in semiconductor fabrication, reflecting the elevated market risk premium required by institutional allocators under trade policy uncertainty. The 1.37 beta reflects the cyclical amplification of semiconductor capital equipment cycles and the foundry's exposure to enterprise computing investment swings.

Applying a 9.79% discount rate against cash flows generated primarily in capital-intensive fabrication assets ensures that our valuation does not understate the cost of capital during major node transitions. The framework discounts projected unlevered cash flows through FY2030 before calculating a terminal value based on a 3.0% perpetual growth rate in the base case, capturing the long-term steady-state expansion of global silicon demand without relying on unrealistic terminal multiples.

FCF margin normalization is the engine

The primary driver of our divergence from the prevailing market price is the trajectory of free cash flow conversion. While the company achieved a 26.1% FCF margin in FY2025 on $992.4B of free cash flow, the step-up in capital intensity in FY2026, guided to $60-64 billion, significantly depresses near-term cash conversion. As capital expenditures absorption occurs alongside N2 depreciation, free cash flow generation is constrained relative to top-line acceleration.

Our model bridges terminal FCF margins across the three scenarios: 20.9% in the Bear case, 23.0% in the Base case, and 23.5% in the Bull case. These assumptions reflect the structural reality that overseas fabs carry higher operating cost baselines than Taiwan facilities, while leading-edge node progression requires continuous, multi-billion-dollar outlays for lithography and advanced packaging equipment, preventing structural FCF margins from expanding beyond previous peak cycles.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**9.79% (2026-10-29)**
Terminal growth	**3.0%**
Equity beta	**1.37**
Diluted shares	**5.19 billion**

HOLD

over $284.15 blended target

Competitive moat and concentration risk

We rate the moat as Wide and we view the market structure as Near-monopoly at leading edge; oligopoly overall foundry. We assign a premium of +30% to sector median multiples on moat strength alone.

- foundry scale plus process lead, 3nm at sustained full utilization, 2nm ramping to 8% of revenue in a single quarter (Q3 2026, TechTimes October 15, 2026), and CoWoS advanced packaging with spillover demand.
- IDC (2026), TSMC raised wafer pricing by more than 5% while raising its 3nm monthly capacity target to 165,000 wafers and CoWoS to 125,000 wafers, on sustained full utilization, raising price into full utilization is the cleanest pricing-power test in this batch. Corroborating, 2nm wafers at ~$30,000, roughly 50% above 3nm pricing, with demand unabated (TechTimes, October 15, 2026).
- The weakest element is concentration, high-performance computing (AI accelerators) was 68% of Q3 2026 revenue (+22% sequentially), and essentially all leading-edge capacity sits in Taiwan, so geopolitical and single-cycle concentration cut against the top of the band.

On a multiple cross-check, the justified range is 30.0-33.8x / 22.9-25.8x. The cross-check value is $224.0-$252.3 (pt $242.6), on a base of $9.78/ADR (49.328% normalized EBIT × FY2025 revenue $122.42bn × 0.84 ÷ 5,186.5M ADRs). The gap verdict is SHRINKS, in part small part ~11% at point (2-16% range).

This figure uses the normalized base from the October 27 recheck, not the original boom-year base.

We have no disclosed backlog figure for this company, so backlog gives no support to the growth path here.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	8 points
single supplier, single fab, or single geography exposure	8 points
platform, channel, or distribution dependency	0 points
customer insourcing risk	4 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 37 percent. Shown side by side with the moat premium or discount, the net stance is +10 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all (corrected from batch arithmetic)

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on High-Performance Computing Demand and Advanced Node Transition Velocity HOLD

The primary revenue catalyst for Taiwan Semiconductor Manufacturing Co is the secular demand for high-performance computing platforms, which have expanded to represent 68% of total revenue as of Q3 FY2026, up from 66% in Q2, 61% in Q1, and 55% in Q4 FY2025. This surge is underpinned by fabless customers including NVIDIA, AMD, Apple, and Broadcom procuring cutting-edge silicon for artificial intelligence accelerators. Sequentially, HPC revenue

expanded 22% in Q3 FY2026 and 20% in Q2 FY2026. This dynamic led management in Q1 FY2026 to increase the 2024-2029 AI accelerator revenue CAGR outlook from 45% to 54-56%, providing multi-year visibility for enterprise computing demand.

The second foundational driver is the rapid customer migration toward advanced fabrication nodes, defined as 7-nanometer and below, which constituted 79% of total wafer revenue in Q3 FY2026. Within this leading-edge envelope, the transition to the 2-nanometer node has progressed at an unprecedented rate, scaling from 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026. Concurrently, the 3-nanometer node represented 25% of wafer revenue in Q1 FY2026 and 28% in Q4 FY2025, alongside 5nm at 35%. Management's Q4 FY2026 revenue guidance of $52.3-54.3 billion, featuring a $53.3 billion midpoint, signals strong sequential growth of approximately 14%, confirming sustained backlog for advanced nodes.

Conversely, smartphone platform exposure has declined as a proportion of total revenue, contracting from 32% in Q4 FY2025 to 22% in Q2 FY2026 and 21% in Q3 FY2026. While the smartphone segment maintains high absolute wafer volumes, its cyclicality and slower secular growth mean high-performance computing will dictate corporate operating leverage. The capital allocation program has responded accordingly, with full-year 2026 capex guided upward to $60-64 billion, following nine-month actual spend of $40.64 billion, to support both front-end N2 capacity and CoWoS packaging facilities.

What to monitor

- **2-Nanometer Node Gross Margin Dilution** Monitor the margin impact of early N2 depreciation, which brought gross margin down from 67.7% in Q2 FY2026 to 66.4% in Q3 FY2026, with Q4 FY2026 gross margin guided between 65% and 67%.
- **Execution Against Full-Year 2026 Capex Guidance** Track capital deployment against the raised $60-64 billion capex target, particularly after nine-month outlays reached $40.64 billion, assessing its dampening effect on full-year free cash flow conversion.
- **Working Capital Metrics and Inventory Expansion** Observe inventory days, which increased to 87 days from 76 days a year prior in Q2 FY2026, alongside receivable days rising to 29 from 23, to assess potential supply chain digestion.
- **Geopolitical and Trade Policy Headwinds** Evaluate the operational impact of the 25% Section 232 semiconductor tariffs implemented in January 2026 and ongoing shifts in export controls regarding advanced computing integrated circuits.

+31.6%	59.9%	50.8%	26.1%	$1,703.3B	31.5%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$3,809.1B
Gross Profit	$2,281.3B
EBIT	$1,936.1B
Net Income	$1,697.6B
EBITDA	$2,742.1B
Free Cash Flow	$992.4B
Diluted EPS	$331.25
ROE	31.5%

Risk & moat matrix

The bear case

SHARE EROSION

While TSMC maintains a monopoly in sub-5nm logic, any technological progress by competing foundries could threaten pricing leverage on future node iterations. However, high customer switching costs among core fabless designers currently mitigate

M&A DIGESTION

TSMC focuses purely on organic fab expansion rather than large-scale acquisitions, avoiding direct integration pitfalls. Capital risk is instead concentrated in the execution and scaling of massive greenfield fabrication facilities globally.

REGULATORY AND PRICING SCRUTINY

The imposition of 25% Section 232 semiconductor tariffs in January 2026 highlights intensifying cross-border trade friction. Furthermore, export control postures on advanced computing integrated circuits to China remain fluid, creating recurring geopolitical overhangs.

TECHNOLOGY DISRUPTION

The rapid shift to N2 introduces substantial execution risk and equipment depreciation that already compressed gross margin to 66.4% in Q3 FY2026. Any unexpected yield difficulties at 2nm could disrupt the planned delivery schedule for key HPC customers.

REINVESTMENT TREADMILL

Capital intensity has escalated dramatically, with FY2026 capex guided to $60-64 billion following $40.64 billion spent across the first nine months. Overseas fab expansions introduce structural cost premiums relative to Taiwan, permanently increasing the capital required to sustain growth.

WACC AND DURATION SENSITIVITY

At our 9.79% discount rate, the equity valuation is highly sensitive to long-dated cash flow timing and terminal growth assumptions. The market's spot price of $432.75 implies cash flow compounding rates far beyond what capital-heavy foundries

immediate market share loss.

historically deliver.

The strengths

OLIGOPOLY POSITION

TSMC operates as the monopoly supplier of sub-5nm logic with CoWoS advanced packaging, securing critical supply lines for premier fabless designers. Advanced nodes at 7nm and below represent 79% of wafer revenue, leaving peers far behind in commercial volume.

FCF COMPOUNDING

The enterprise generated $992.4B in free cash flow in FY2025, supported by exceptional operating margins that reached 60.3% in Q2 FY2026. High operating cash flow generation provides substantial self-funding capability even during elevated capex cycles.

GEOGRAPHIC DURABILITY

TSMC's established manufacturing base in Taiwan delivers unmatched supply chain clustering and engineering efficiency. While overseas expansions carry cost premiums, they diversify physical operational footprints to meet client procurement criteria.

PIPELINE OPTIONALITY

The company's raised AI accelerator revenue CAGR outlook of 54-56% through 2029 demonstrates robust structural demand across multiple platform generations. Customer adoption velocity at 2nm, scaling from 3% to 8% of wafer revenue in one quarter, confirms rapid technological absorption.

The bottom line on risk and reward

Taiwan Semiconductor Manufacturing Co remains an execution powerhouse with operational metrics that lead the global semiconductor industry. The company's ability to generate NT$763.06 billion in quarterly net profit while growing revenue by 41.2% year over year in Q3 FY2026 demonstrates unmatched operational leverage. However, the investment proposition is fundamentally constrained by valuation. The market capitalization of $2,244.5B at the reference price of $432.75 prices in flawless execution while ignoring the cash flow friction created by $60-64 billion in annual capital expenditures.

Our quantitative valuation model, employing a 9.79% WACC and an 11.6% base revenue CAGR, yields a probability-weighted blended target price of $284.15. This indicates that current trading levels represent a -34.3% downside risk. While long-term enterprise demand for artificial intelligence silicon is undeniable, current prices discount multi-year cash flow delivery that leaves no margin of safety for depreciation drag, working capital expansion, or trade policy disruption. Consequently, we maintain a Hold rating with pronounced valuation concerns.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

Leading-Edge Logic and CoWoS Advanced Packaging: The AI Supercycle Monopolist

Taiwan Semiconductor Manufacturing Co's core operational franchise centers on leading-edge logic fabrication, where nodes at 7-nanometer and below generate 79% of wafer revenue. The integration of front-end logic fabrication with CoWoS advanced packaging has established the company as the sole manufacturing source for critical hyperscaler AI accelerators designed by fabless partners such as NVIDIA, AMD, Apple, and Broadcom. By providing turnkey fabrication and packaging solutions, the company captures the majority of economic rents generated within advanced logic manufacturing.

The velocity of customer adoption at the leading edge continues to break historical records. The transition to the 2-nanometer node demonstrated unprecedented acceleration, climbing from 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026. Fabless customers are migrating to leading nodes faster than in any previous generation, driven by the compute and energy efficiency demands of artificial intelligence workloads. This demand enabled the high-performance computing platform to expand sequentially by 20% in Q2 FY2026 and 22% in Q3 FY2026, reaching 68% of the corporate platform mix.

However, running this manufacturing franchise requires substantial capital reinvestment that alters corporate cash flow dynamics. The front-loaded capital outlays required for N2 and CoWoS expansions drove capital expenditures to $13.84 billion in Q3 FY2026 alone, bringing nine-month spending to $40.64 billion. Management raised the full-year 2026 capex guidance to $60-64 billion, up sharply from $40.9 billion in 2025. In addition to high capital intensity, early N2 depreciation compressed gross margin to 66.4% in Q3 FY2026, with Q4 gross margin guided between 65% and 67%, down from the Q2 peak of 67.7%.

What the recent record shows

- Record Q3 FY2026 revenue of $46.72 billion (+41.2% YoY) exceeded the $45.8 billion guidance ceiling, accompanied by a Q4 revenue guide of $52.3-54.3 billion.
- The 2-nanometer node ramped rapidly from 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026, while 7nm and smaller nodes reached 79% of total wafer revenue.
- Early N2 depreciation compressed gross margin to 66.4% in Q3 FY2026 from 67.7% in Q2 FY2026, with management guiding Q4 gross margin to 65-67%.
- Capital intensity escalated significantly, with cumulative nine-month FY2026 capex reaching $40.64 billion and full-year 2026 capex targeted at $60-64 billion.

Our read: TSMC's advanced node logic and CoWoS packaging franchise remain technically peerless and commercially indispensable to global technology leaders. However, the sheer capital intensity required to scale 2nm capacity and overseas facilities suppresses free cash flow margins, leaving the stock vulnerable at current valuation levels.